

Flinders University

College of Science and Engineering

EWB Challenge — 2023

Mangrove Erosion Control System and Stabilising Structure for The Barron River

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Contents

Executive Summary 3
Introduction 4
Design Development 5
Design Options 7
Final Design 15
Design Assessment 19
Design Implementation 24
Conclusion 26
Acknowledgments 26

List of Figures

1 *Barron River sub-catchments* 4
2 *Brunswick River mouth in NSW before building a breakwater structure (left) and the river's mouth after building the breakwater showing the effects of limiting erosion (NSW Marine Estate, 2023)* 7
3 *Construction of breakwater mound structure from rocks and rubble (Ur Rahman, 2021)* . . . 7
4 *Excavating mangrove saplings using a shovel (left), and transpiration of excavated saplings (right) in Al Mubarraz Island, United Arab Emirates (Erftemeijer et al., 2021)* 8
5 *Mangrove seedling planted in a recycled metal box (left), and mangrove propagules (ingrown-seedling) planted in milk cartons (right) (Melbourne Water, 2013)* 9
6 *Mangrove seedlings wide implantation example (Melbourne Water, 2013)* 9
7 *Mangrove seedlings growing in fabricated PVC pipe (left), and wide implantation of seedlings sheltered in PVC pipe (right) (Melbourne Water, 2013)* 10
8 *Crib wall made of tree logs in the Guadalupe River in San Jose, California (Cooperative Conservation, 2001)* 11
9 *Concrete crib wall (Gabion 1)* 11
10 *Lattice-pattern wattle structure (left), and braided fascines structure (right) (Evette et al., 2009)* 12
11 *Design consideration matrix comparing solutions based on weighted average calculated considerations scores* 14
12 *Final Design Simplified Functional Diagram* 17
13 *3D Printed Prototype from the designed AutoCAD model* 19
14 *3D Model render of the prototype fences installed on a river bend model while empty* 20

15 *Render of the 3D river bend model terrain points (orange) and fence boundaries(black)* 20
16 *Steady-state CFD modulated water velocity contours applied for the simulation* 21
17 *Steady-state CFD simulation result showing velocity magnitude of water current in cm/s, showing areas were force exerted heavily (red) and weakly(blue)* 21

Executive Summary

Objective: This report proposes an eco-friendly, cost-effective solution for mangrove erosion along the Barron River near Dungarra, Australia, incorporating community engagement strategies respecting the Djabugandji people's cultural norms and traditions.

Background: The Barron River, also known as Djirri Nyundu Nyrrumba by its Traditional Owners, the Djabugandji people, serves as a crucial water source for agricultural, industrial, and residential purposes. The local community's livelihood depends on the river, which has been increasingly impacted by climate change, endangering both the ecosystem and the community's well-being.

Design Process: A collaborative effort among Engineers Without Borders and a team of six authors led to the development of a stabilisation structure, with careful consideration given to the local environment, community involvement, planting methods, and the system's long-term sustainability and maintenance.

Design Selection: The water wattle fences design emerged as the most viable solution based on a decision matrix that factored in cost, sustainability, feasibility, and maintenance. This design offers protection to mangroves against flooding, is easy to maintain by the community, and ensures social and economic benefits.

Assessment Testing: After successful preliminary testing on a 3D-printed prototype and conducting a Computational Fluid Dynamics steady-state simulation, a comparative analysis was conducted on the cost and benefits of wattle fences to other alternatives. The wattle fences, made from sticks of locally available trees, were found to be cost-effective and environmentally friendly.

Implementation Plan: The plan suggests progressive execution with priority given to high erosion zones. Building materials will be sourced locally, with minimal ecological risk. Most expenses are expected to be labour costs. A comprehensive framework for governance and organisation will be implemented for oversight and sustenance of the project. Funding will be sought from governmental bodies, community donations, non-profit organisations, and crowdfunding platforms.

System Maintenance: Regular inspections, trimming invasive vegetation, creating community-based maintenance guidelines, and organising training sessions are all part of the maintenance plan. The aim is to encourage community participation and build a sense of ownership.

Conclusions and Recommendations: The implementation of water wattle fences offers a sustainable, cost-effective, and community-centric solution to mangrove erosion control along the Barron River. The approach integrates the Djabugandji community's values and contributes to the preservation of the river's ecosystem. Further research, testing and simulation, and monitoring are suggested to enhance the long-term effectiveness of the system.

Introduction

Background

The Barron River, known as Djirri Nyundu Nyrrumba by its Traditional Owners, the Djabugandji people, is an essential waterway in far north Queensland, Australia. It spans 165km and flows into the Coral Sea at Cairns, providing water for agriculture, industry, and residents. The Djabugandji people have a deep cultural and spiritual connection to the river, relying on it for their livelihoods, including farming, fishing, and supporting local tourism.

The Barron River catchment is located on the Queensland wet tropical coast and covers a relatively large area of 219,000 hectares *(Russell et al., 2000)*. It encompasses population centres such as Cairns, Kuranda, Mareeba, Walkamin, Atherton, Yungaburra, and Tolga. The catchment also includes Half Moon Creek, a smaller coastal catchment north of the Barron River, with the Macalister Range forming its western boundary. Within the catchment, there are several local government areas with approximate populations, including Cairns (4,700 in Barron River catchment suburbs), Eacham (6,300), Mareeba (18,200), Mulgrave (8,100), and Atherton (10,100) *(Carr, 1994, Cook et al., 1998)*. The suburb of Yorkeys Knob, located in the Half Moon Creek catchment, has an estimated population of 2,400 *(Cook et al., 1998)*.

The headwaters of the Barron River can be found in the southern part of the catchment, within the Mount Hypipamee National Park. The river traverses the undulating dairying country of the upper Atherton Tablelands before flowing into Lake Tinaroo. Lake Tinaroo has a storage capacity of 436.5 gigaliters (GL), covers an area of 33.7 square kilometres, and has a shoreline length of about 209 kilometres. Townships in the upper catchment area include Atherton, Kairi, and Yungaburra. The river then continues northward, passing through the population centre of Mareeba before turning east towards the coast. Near the township of Kuranda, the Barron River cascades approximately 300 meters down the Barron Falls onto the coastal plain. From there, it meanders through forests, sugar cane farms, urban areas, and mangroves before discharging into the Coral Sea. Another significant water storage facility, Copperlode Dam (impounding Lake Morris), is located in the upper reaches of Freshwater Creek. It has a storage capacity of 44.5 GL, covers an area of 3.1 square kilometres, and has a shoreline length of about 39 kilometres*(Russell et al., 2000)*.

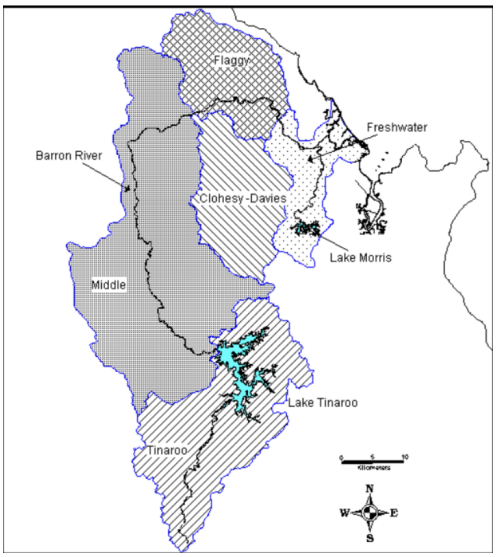


Figure 1: *Barron River sub-catchments*

As for Half Moon Creek, it covers an approximate area of 954 hectares and is situated at the northern end of the Barron River delta. It primarily functions as a tidal creek, draining into the Coral Sea at Yorkeys Knob. The creek plays a role in draining extensive tidal wetlands, with the predominant land use in the catchment being sugar cane farming.

The water infrastructure within the Barron River catchment is highly regulated, with five impoundments, including two major storages. Tinaroo Falls Dam, located on the upper Barron River, impounds Lake Tinaroo, while Copperlode Dam impounds Lake Morris on Freshwater Creek. Tinaroo Falls Dam supplies water to the Mareeba-Dimbulah Irrigation Area (MDIA) through a network of irrigation channels, siphons, and balancing storages. Additionally, channel systems have been constructed in some areas to facilitate regulated flows of irrigation water in natural conditions.

The Barron River is facing an erosion problem, which has been intensified by climate change. This erosion jeopardises the river's ecosystem and the well-being of the local community. To address this challenge, planting mangrove trees can provide an economically viable solution for safeguarding riverbanks from erosion while also promoting biodiversity by establishing fresh ecological niches and natural carbon neutralisers *(EWB Challenge Brief, 2023)* which has already been attempted by the Dawul Wuru Abroginal Corporation. However, planting mangrove seedlings was unsuccessful due to them being washed away by water currents, hence using a sustainable and cost-effective solution is needed to stabilise mangrove seedlings on riverbanks and protect them from being washed away to implement a sustainable erosion control system.

The community's need is for a solution that aligns with their cultural values and respects their traditions. They seek a solution that is environmentally friendly, economically viable, and socially and culturally appropriate. The solution should actively involve the Djabugandji community, ensuring their participation and ownership in the design and implementation process. Long-term sustainability, including maintenance and preservation guidelines and plan, is also essential to ensure the ongoing effectiveness of the erosion control measures.

Overall, the community requires a sustainable erosion control solution that safeguards the riverbanks, preserves the mangrove vegetation, and maintains the ecological balance of the Barron River. This solution should be tailored to the specific needs and values of the Djabugandji community while considering the economic, social, cultural, and environmental aspects of the region.

Problem Definition

The problem that the project aims to solve is the erosion of riverbanks along the Barron River near Dungarra, Australia, which poses a threat to the ecosystem and the well-being of the local community, especially the Djabugandji people. The design challenge is to develop a sustainable and cost-effective solution to stabilise mangrove seedlings on the riverbanks and mitigate erosion caused by floods and river currents. The problem statement can be defined as follows: ***"How might we design a sustainable and cost-effective solution to stabilise mangrove seedlings on riverbanks for erosion control along the Barron River near Dungarra, Australia?***"

Design Development

Design Considerations

The design considerations were developed to tackle considered design challenges such as robust stabilisation structures, efficient planting methods, minimal environmental impact, community involvement,

and long-term maintenance and sustainability.

*

- **Stabilisation structures** Develop robust structures that can effectively stabilise mangrove seedlings on riverbanks for erosion control. This consideration focuses on the structural integrity of the design, ensuring it can withstand floods and river currents.
- **Planting methods** Implement efficient and sustainable methods for planting mangroves along the riverbanks. Consider ecological aspects such as the selection of native mangrove species and the durability of planting techniques to enhance the survival and growth of the seedlings.
- **Environmental impact** Evaluate the potential impact on the local environment, including the river ecosystem and wildlife. Consider the ecological consequences of building the structure and implementing the erosion control solution, and aim to minimise any adverse effects.
- **Community involvement** Involve the local Djabugandji community in the design process to ensure their economic, social, and cultural needs are considered. Engage community members in decision-making, economic aspects, and cultural integration to foster ownership and a sense of empowerment.
- **Maintenance and sustainability** Consider the long-term maintenance and sustainability of the project. Evaluate the durability of the chosen design, and its maintenance needs, and develop comprehensive guidelines for routine inspections, repairs, and replacements with the aid of implementing a comprehensive governing framework.

Metrics for evaluation

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- **Structural stability:** Assess the strength and resilience of the stabilisation structures to withstand flood and current forces. Measure the performance and longevity of the structures under different conditions.
- **Plant survival rate:** Track the success rate of mangrove seedling planting methods by monitoring their survival and growth over time. Measure the establishment and development of mangrove vegetation.
- **Environmental impact:** assessment: Conduct an assessment of the project's environmental impact, considering factors such as habitat disturbance, species diversity, and water quality.
- **Community engagement:** Measure the level of community participation and engagement throughout the project. Assess the extent to which the design considers the community's economic, social, and cultural needs.
- **Maintenance effectiveness:** Evaluate the effectiveness of the maintenance guidelines in identifying and addressing wear, damage, or potential hazards. Measure the timeliness and quality of repairs and replacements.

By considering these design considerations and utilising appropriate metrics for evaluation, the project can develop a sustainable and effective erosion control solution that aligns with the needs of the Djabugandji community and the Barron River ecosystem.

Design Options

Breakwater structure

A breakwater is a structure implemented parallel or close to shores, harbours, lakes, and river mouths, that "breaks" waves to protect shores from damaging currents, waves and storm surges. Breakwaters are barrier wall structures usually made of rocks and rubble, concrete, and steel, supplying a rigid and robust physical barrier that dissipates the energy of incoming waves and currents by reducing their intensity and preventing their significant effects from eroding the shoreline, infrastructure, and ecosystems on inhibiting shorelines *(Climate ADAPT, 2016)*. Breakaways are ubiquitously used in mainly coastal shorelines and in marine sites where a strong and highly resilient barrier is required to counteract waves and currents caused by external factors such as oceanic and sea currents. The planning and construction of breakaways need specialised engineering and construction techniques and incur a high cost of labour, materials and logistics, planning and construction.



Figure 2: *Brunswick River mouth in NSW before building a breakwater structure (left) and the river's mouth after building the breakwater showing the effects of limiting erosion (NSW Marine Estate, 2023)*

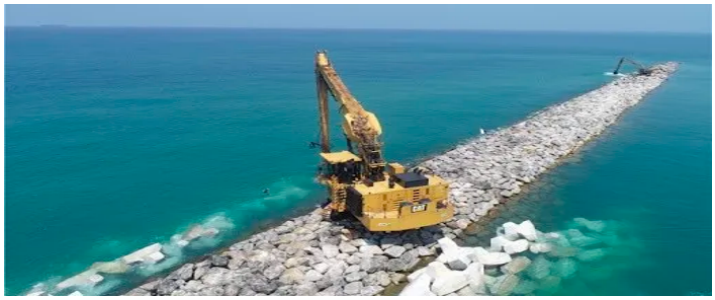


Figure 3: *Construction of breakwater mound structure from rocks and rubble (Ur Rahman, 2021)*

* Advantages

- Highly effective in mitigating impacts from waves and currents due to its high resilience against water forces.
- Particularly sustainable and durable. Building materials can be sourced from local and sustainable sources, and the structure rarely requires maintenance.

* Disadvantages

- Breakwaters are mainly a solution to mitigate and counteract waves and currents originating from external marine sources do not account for other external factors causing currents and floods such as rainfall and other internal factors driving currents in the river and also do not offer direct protection to shelter mangrove seedlings from direct forces and wash away. Breakaways can be efficient solutions to protect river mouths from sea currents.
- The planning and construction of breakaways are costly and labour-intensive and require specialised expertise, labour, and equipment, which do not contribute to advancing the local community's economy.

Relocation of salvaged half-grown mangrove plants

Relocating mature mangrove plants in order to ensure that the mangroves develop long and strong roots capable of withstanding the flooding and shocks from river waves, growing them beforehand is a viable solution. Once they have reached a mature stage of growth, they can then be replanted along the river at the main site. This approach allows the mangroves to establish a sturdy foundation and better adapt to their environment *(Erftemeijer et al., 2021)*.



Figure 4: *Excavating mangrove saplings using a shovel (left), and transpiration of excavated saplings (right) in Al Mubarraz Island, United Arab Emirates (Erftemeijer et al., 2021)*

* Advantages

- The roots will remain firmly anchored in the ground, withstanding the impact of water waves from the river and effectively mitigating the risk of flooding.

- Replanting mangroves can also be easy to maintain and low in cost if there were no materials involved.

* Disadvantages

- The plants are better suited to be grown in the same environment it was grown in as the change in the amount of water and salt in the soil affects. They were adapted to support how well the trees can grow and are better left grown where they are grown.
- Requires larger and more costly means of transportation, logistics, and care.

Revegetation of seedlings in protective shelters

Revegetation of nursery-planted mangrove seedlings which later are placed in a protective casing made from low-cost and recycled materials readily available in local communities such as paper milk cartons, metal cans, PVC pipes, and other creative recycled materials *(Melbourne Water, 2013)*. The protective casing serves to shelter and house the seedlings to sustain their growth and maturation until they are able to withstand currents and flooding from being washed away.



Figure 5: *Mangrove seedling planted in a recycled metal box (left), and mangrove propagules (ingrown-seedling) planted in milk cartons (right) (Melbourne Water, 2013)*



Figure 6: *Mangrove seedlings wide implantation example (Melbourne Water, 2013)*



Figure 7: *Mangrove seedlings growing in fabricated PVC pipe (left), and wide implantation of seedlings sheltered in PVC pipe (right) (Melbourne Water, 2013)*

* Advantages

- A low-cost, environmentally and economically sustainable solution sourced by readily available recycled and biodegradable materials, and more conventional and easily applicable compared to other revegetation techniques.
- Need minimal maintenance, especially once mangroves are grown and matured to sustain environmental factors.

* Disadvantages

- Do not offer resilient durability and strength against rough currents and extreme environmental forces.

River crib walls

The employment of river crib walls is a commonly used technique to reduce river erosion and, in this particular case, they can be constructed to prevent river floods from reaching and washing away plants. Crib walls are typically built using materials such as concrete, precast blocks, logs, and timber. However, the implementation of crib walls can be quite costly due to the need for skilled engineering expertise, intricate planning, and ongoing specialised maintenance and labour.



Figure 8: *Crib wall made of tree logs in the Guadalupe River in San Jose, California (Cooperative Conservation, 2001)*



Figure 9: *Concrete crib wall (Gabion 1)*

* Advantages

- Crib walls redirect the flow of water in the river, ensuring a single directed path, and are capable of withstanding the strong forces of currents and waves.
- They effectively block floodwaters from reaching the mangroves, allowing them to grow and thrive without being washed away.

* Disadvantages

- The construction of crib walls can be considerably expensive and technically challenging, with each project's structure being specific to the site, making it difficult to replicate in other locations.
- Modifying the natural course and structure of the river through crib walls requires significant intervention, which can impact the existing ecosystem and disrupt its ecological balance.

Water channelling

Water channelling for rivers refers to the construction of artificial channels or canals that are designed to redirect or modify the flow of a river. Channels can be made in many ways, such as Flood Relief Channels and Channel Straightening. Also, by digging in the ground to channel water and keep the water moving to control the flooding.

* Advantages

- Reduces the amount of water that seeps out of the river.
- It helps to avoid flooding and improve water flow after heavy rain and the large volume of water influx.

* Disadvantages

- Channelling can significantly increase water flooding velocities, exacerbating erosion within the river and posing a greater risk to its stability and integrity.
- These channels are highly susceptible to flooding when river levels rise, further exacerbating the potential for damage and erosion.
- The construction and excavation process for creating these channels necessitate the use of specialised machinery, resulting in significant costs associated with their implementation.

Water Wattle Fences

This design involves the construction of wooden stick fences supported by wooden fascines firmly embedded in the ground. Water wattle fences are positioned alongside the river to counteract the detrimental impacts of powerful waves and currents, which can cause erosion and flooding, leading to the loss of mangrove seedlings and disruptions to the river's natural flow.

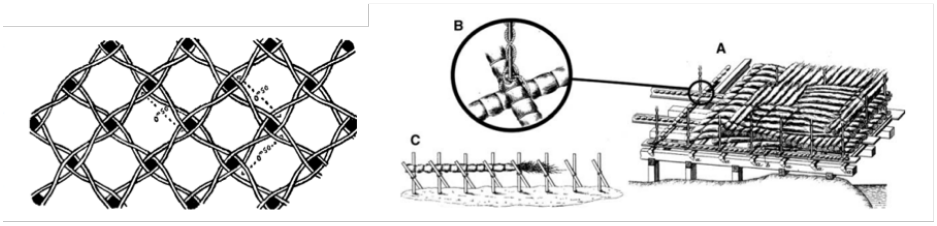


Figure 10: *Lattice-pattern wattle structure (left), and braided fascines structure (right) (Evette et al., 2009)*

* Advantages

This approach offers several advantages compared to alternative methods:

- Native plants are better adapted to thrive in their original environment, as they have already acclimated to the specific water and soil conditions. This natural adaptation supports optimal tree growth and ensures their suitability for the local habitat.

- Low-cost and can be locally made and maintained.
- The water wattle fences act as a protective barrier, effectively mitigating erosion and flooding, safeguarding the riverbanks and preserving the stability of the surrounding ecosystem.

* Disadvantages

Despite its benefits, this approach does have certain limitations to consider:

- Regular maintenance and monitoring are necessary to ensure the ongoing stability and durability of the water wattle fences, as natural wear and tear can occur over time. Proper upkeep is vital for maintaining their long-term effectiveness in preserving the integrity of the river ecosystem.
- Following severe weather events, such as storms, it is essential to conduct prompt inspections of the fences. They are susceptible to damage from strong winds, debris, and lightning strikes, which could compromise their effectiveness in protecting the riverbanks.

Design Matrix Analysis

The design matrix was created to evaluate the proposed design options for stabilising mangrove seedlings on riverbanks for erosion control along the Barron River near Dungarra, Australia. The matrix considered important design considerations such as cost, sustainability, feasibility, and maintenance. Each design option was scored based on these criteria, and the option with the highest score was chosen as the final design. The design options evaluated were:

1. Breakwater structure
2. Relocation of salvaged half-grown mangrove plants
3. Revegetation of seedlings in protective shelters
4. River crib wall
5. Water channelling
6. Water wattle fences

Design Considrations Matrix										
		Cost		Sustainbility		Feasibility		Maintainance		Total
	Weight	2		3		4		5		
Sloutions	Breakwater Structure	2	4	5	15	2	8	5	25	52
	Replanting Grown Plants	3	6	3	9	3	12	3	15	42
	Contained Seedlings	5	10	2	6	5	20	1	5	41
	Water Wattle Fences	4	8	4	12	4	16	4	20	56
	Water Channelling	4	8	3	9	3	12	3	15	44
	River Crib Wall	1	2	5	15	1	4	5	25	46
		Raw Score	Weighted Score	Raw Score	Weighted Score	Raw Score	Weighted Score	Raw Score	Weighted Score	Total Weighted Score
	Weighting Criteria	Does not address design consideration ← 1 2 3 4 5 → Addresses design consideration well								

Figure 11: *Design consideration matrix comparing solutions based on weighted average calculated considerations scores*

After conducting research and scoring each design option, the water wattle fence design emerged as the highest-scoring option on the matrix. This design was chosen as the final design for the following reasons:

1. Protection from flooding: The water wattle fences design proved effective in blocking water during testing, indicating its ability to protect vulnerable seedlings and immature plants from flooding.

2. Community engagement: The design focuses on being easy to maintain by the community and incorporates engagement of the local Djabugandji community. This aspect aligns with the community-centric approach and promotes social and economic benefits for the community.

3. Cost-effectiveness: The wattle fences, made from sticks of raw resourced trees, have a relatively low cost compared to other alternatives. This factor ensures the project's affordability and sustainability.

4. Environmental compatibility: The wattle fences are made of natural materials and pose minimal threats to the environment. They can be constructed using widely available, non-endangered native trees, further promoting ecological sustainability.

5. Adaptability and repairability: The wattle fences design offers ease of repair and adaptability. The design can be customised based on the river's specific needs and can be replaced or repaired using traditional techniques and locally available resources.

   Based on these factors, the water wattle fences design emerged as the most suitable option for stabilising mangrove seedlings on riverbanks along the Barron River. Its effectiveness, cost-effectiveness, community engagement, and environmental compatibility make it a well-rounded choice for the erosion control project.

Final Design Selection

After evaluating the design options and considering the design matrix scores, the final design selected for the project is the water wattle fences. This design best aligns with the project's goal of

developing a sustainable and cost-effective solution for stabilising mangrove seedlings on riverbanks for erosion control.

The water wattle fences design offers several advantages, including its ability to protect mangroves from flooding, its ease of maintenance by the community, and its incorporation of community engagement and social-economic benefits. Furthermore, the use of natural materials in the wattle fences minimises the environmental impact, and its low cost makes it an affordable and sustainable option.

By selecting the water wattle fences as the final design, the project aims to ensure the preservation of the Barron River ecosystem, protect natural habitats, and meet the needs and values of the local Djabugandji community. Continuous research, monitoring, and community involvement will enhance the long-term effectiveness and adaptability of the erosion control system.

Final Design

Overview of Final Design

The final design selected for the erosion control project along the Barron River near Dungarra, Australia is the water wattle fences. This design aims to provide a sustainable and cost-effective solution to stabilise mangrove seedlings on riverbanks, mitigating erosion caused by floods and river currents. The water wattle fence design was chosen based on its effectiveness in protecting mangroves from flooding, its ease of maintenance by the community, and its incorporation of community engagement and social-economic benefits.

*** Technical Aspects**

- Stabilisation Structures: The water wattle fences are robust structures designed to stabilise mangrove seedlings on riverbanks. They consist of wooden wattle fences constructed from widely available, non-endangered native trees such as Blackbutt and Silky Oak. These fences are erected along the riverbanks to create a barrier that prevents the washing away of seedlings due to water currents and floods.
- Construction Materials: The wattle fences are made from natural materials, minimising their environmental impact. The primary construction materials include wooden ground posts, rope or twine for binding the wattle branches, and wire for added stability. These materials can be sourced from raw natural resources or recycled materials, reducing costs and promoting sustainability.
- Implementation Strategy: The implementation of the water wattle fences will be carried out in progressive stages, prioritising areas with high erosion. The project will involve local indigenous labour and community participation, promoting a sense of ownership and cultural integration. Training sessions will be organised in cooperation with indigenous community members, conducted in both English and the Djabugay language, to ensure effective communication and community engagement.
- Maintenance and Repair: The maintenance of the water wattle fences will involve routine walking inspections by designated members of the Barron River community or volunteers. Any wear, damage, or potential hazards caused by natural factors will be identified and repaired promptly. Maintenance guidelines will be developed based on traditional techniques

and practices, using known available resources and parts within the local community. The guidelines will cover common repairs and include instructions for fabricating new wattle fences if needed.

- Environmental considerations: The design of the water wattle fences takes into account environmental considerations. The use of locally sourced non-endangered tree species ensures the preservation of local biodiversity. These species are abundant within the local area's vicinity and form non-invasive ecological relationships within the ecosystem. By utilising natural materials and implementing sustainable construction practices, the project minimises its ecological footprint and promotes the overall health of the Barron River environment.

- Cultural Considerations: The water wattle fences design respects and incorporate the cultural considerations of the Djabugandji community. The involvement of local indigenous labour and community participation ensures cultural integration and a sense of ownership. The training sessions conducted in both English and the Djabugay language facilitate effective communication and engagement, respecting the cultural heritage and traditions of the community. The design acknowledges the significance of the Barron River to the community's identity and aims to preserve and enhance their cultural connection to the river.

- Economical Considerations: The water wattle fence design offers economic benefits and cost-effectiveness compared to alternative options. The materials required for construction can be sourced locally, minimising transportation costs. By involving local indigenous labour, the project creates employment opportunities within the community, contributing to their socio-economic well-being. The use of natural and recycled materials also helps reduce costs. Additionally, the project can secure funding through petitions for governmental support, community donations, non-profit organisations, and crowdfunding platforms. This diversified funding approach ensures the sustainability of the project and covers future maintenance, administrative, and organisational costs.

- Cost and Funding: The water wattle fences design offers cost-effectiveness compared to alternative options. The materials required for construction can be sourced locally, minimising costs. Labour costs, particularly local indigenous labour, will be the highest expense. Funding can be obtained through petitions for governmental support, community donations, non-profit organisations, and crowdfunding platforms. This diversified funding approach will help sustain the project and cover future maintenance, administrative, and organisational costs.

By implementing the water wattle fences design, the project aims to provide an environmentally compatible solution that integrates the needs and values of the Djabugandji community. The design's technical aspects ensure the robustness and durability of the erosion control system while promoting community engagement, cultural preservation, and long-term sustainability.

Functional Diagram

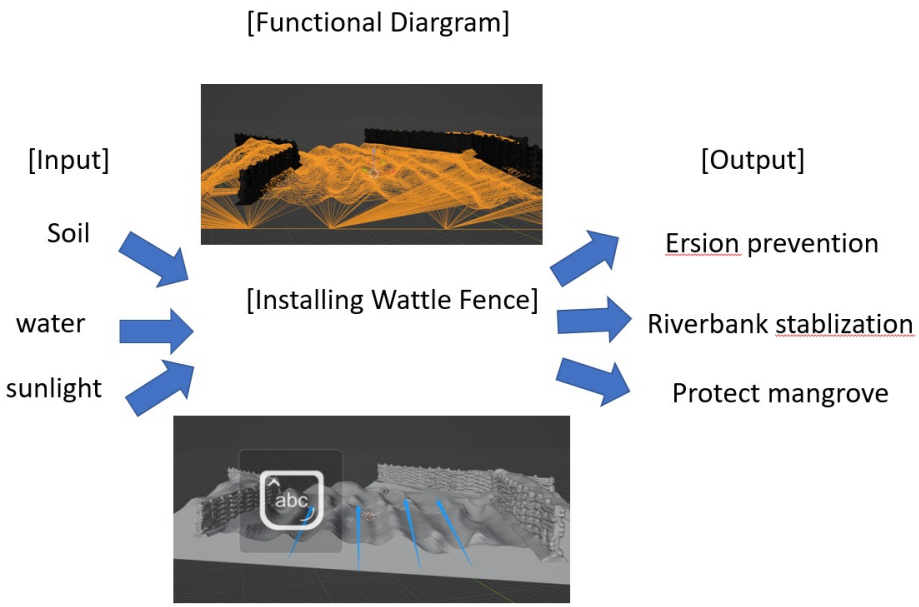


Figure 12: *Final Design Simplified Functional Diagram*

Technical Specifications Guide

This technical guide provides detailed instructions for the construction of water wattle fences, The fences are made from locally available materials and the project aims for the approach to be cost-effective, sustainable, and culturally integrated.

Materials Required:

- Wooden wattle fence posts (made from non-endangered native trees such as Blackbutt, Oak, or Hazel)
- Rope (purchased or made from bark)
- Twine (purchased or made from vines or grasses)
- Wire (purchased or made from recycled materials)
- Tools (hammers, saws, nails)
- Measuring tape or ruler
- Personal protective equipment
- (PPE) such as gloves and safety glasses

Construction:

This is a brief introduction to construction steps using common practices used by farmers that relate to old traditional ways of building wattle fences throughout history, which were found on online websites *(Forgotten Way Farms, 2016)*:

1. Preparation: Identify the areas along the riverbanks with high erosion and prioritise those for construction. Gather all the necessary materials and tools at the construction site. Ensure the availability of adequate manpower for the construction process.

2. Setting up the Fence Posts:

   Measure the desired spacing between the fence posts (e.g., 2 meters) using a measuring tape or ruler. Dig holes for the fence posts along the riverbank, ensuring they are deep enough to provide stability and withstand environmental conditions. Place the wooden wattle fence posts vertically in the holes, ensuring they are securely positioned.

3. Attaching the Horizontal Wattle: Cut and collect flexible branches or sticks from non-endangered native trees. Attach one end of a branch to the first fence post at a suitable height above the ground, using rope or twine. Weave the branch horizontally between adjacent fence posts, ensuring a tight and secure fit. Repeat this process for each horizontal layer of wattle until the desired height is achieved.

4. Strengthening the Wattle Fence:

   Use additional branches or sticks to create diagonal supports between the horizontal layers of the wattle. Attach these diagonal supports to the fence posts using rope or twine, forming a lattice-like pattern. This strengthening technique enhances the stability and durability of the wattle fence, especially during floods and strong river currents.

5. Securing the Wattle Fence:

   Use wire to wrap around the fence posts and the woven branches at regular intervals. This wire wrapping provides additional support and helps maintain the integrity of the wattle fence.

6. Quality Assurance and Inspection:

   Inspect the constructed wattle fence thoroughly, checking for any loose or weak sections. Ensure that the fence is securely anchored to the ground and can withstand environmental forces. Make any necessary adjustments or repairs to ensure the fence is structurally sound.

Maintenance Guidelines:

To ensure the long-term effectiveness of the water wattle fences and the mangrove erosion control system, the following maintenance guidelines should be followed:

Routine Inspections: Conduct regular walking inspections of the wattle fences to identify wear, damage, or potential hazards caused by natural factors. Repair any damage promptly to maintain the integrity of the fences.

Vegetation Management: Trim any invasive vegetation in close proximity to the wattle fences to prevent overgrowth and root heaving that could damage the infrastructure.

Maintenance Training and Guidelines: Organise regular training sessions on the maintenance of the erosion control system and water wattle fences. Develop maintenance guidelines based on traditional techniques and practices, using locally available resources and parts. Include instructions on how to fabricate new wattle fences to replace any defective or damaged fences.

Design Assessment

Preliminary Realisation of Prototype

The prototype of the water wattle fence was successfully created using PLA plastic through 3D printing. To evaluate its effectiveness, in this preliminary evaluation, one side of the fence was filled with water to assess its ability to block water flow. Simultaneously, soil was placed on the other side to measure the displacement of soil caused by the water. This test aimed to gauge both the fence's capacity to impede water movement and its impact on soil stability. The test was an attempt to mimic the test in the study conducted by *(Krishna Prasad et al., 2016)* where the efficacy of their greyones installation to mitigate erosion was measured by the displaced amount of the soil in the banks.



Figure 13: *3D Printed Prototype from the designed AutoCAD model*

However, further analysis revealed that for practical implementation along the river, a longer fence would be necessary, potentially requiring increased height depending on its specific location, with more variable velocity conditions of water flow.

It was preliminary concluded that the wattle fence could provide sufficient robustness for this purpose.

Computational Flow Dynamics Modelling

To further obtain a comprehensive and realistic assessment of the water wattle fence prototype's performance, computational fluid dynamics (CFD) modeling was utilised to obtain a more comprehensive realistic testing result about the roubustness of the proposed fence built. A 3D model of a river bend was built and the 3D prototype fences of both types (straight and bend) were fitted by the sides of the river using AutoCAD and Blender 3D modelling software.

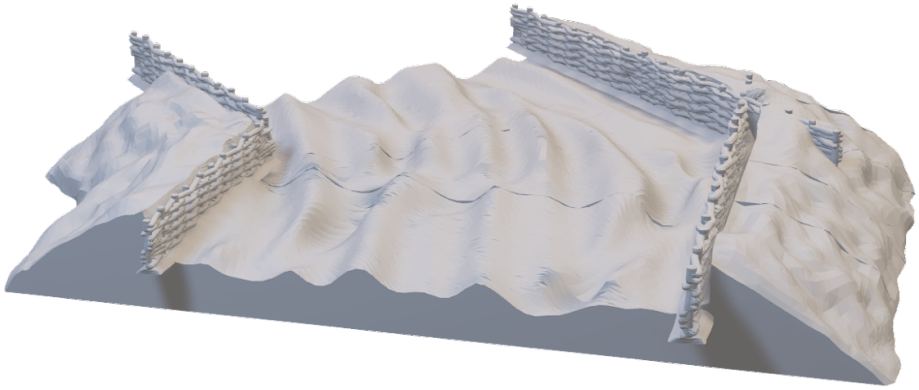


Figure 14: *3D Model render of the prototype fences installed on a river bend model while empty*

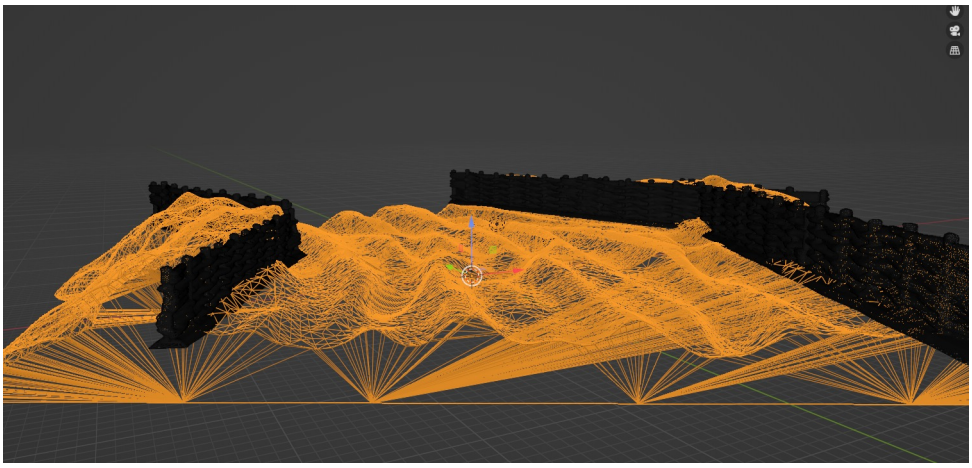


Figure 15: *Render of the 3D river bend model terrain points (orange) and fence boundaries(black)*

The combined 3D model was then imported into AutodeskCFD, where it underwent basic meshing to generate simulation grids, and the material of the fence was assigned to represent the physical properties of the wood. Boundary conditions were designated and set, and a steady-state simulation using water speed velocity was specified at 20 m/s to simulate real-world flow conditions of average water flow velocity conditions upon researching the internet (ChatGPT 4, 2023). The average water flow velocity of the Barron River was unattainable due to the lack of available resources.

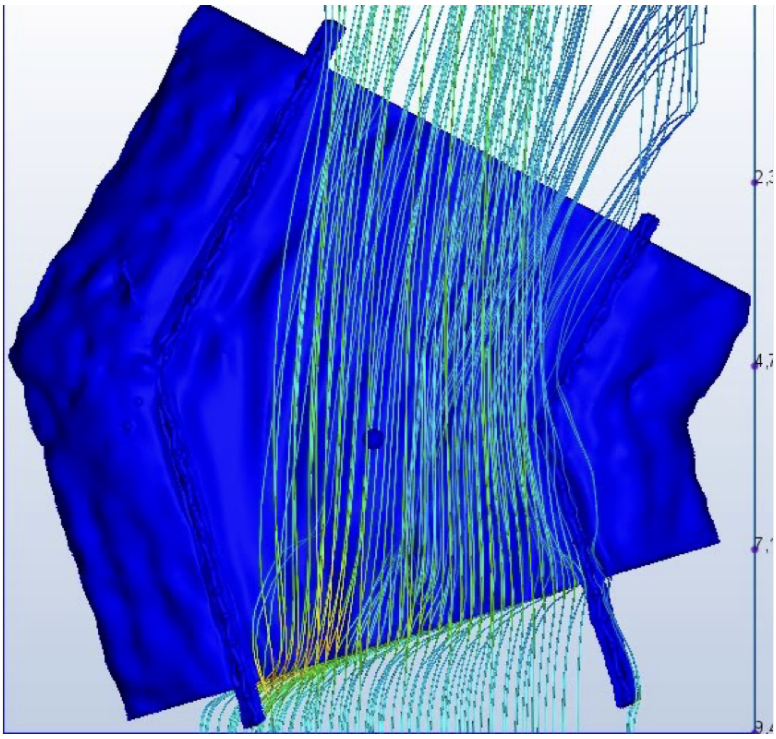


Figure 16: *Steady-state CFD modulated water velocity contours applied for the simulation*

A total of 100 iterations were run in the CFD simulation to ensure that uncertainties were minimised and accurate results were obtained. The simulation provided insights into the behaviour of the water flow and the forces exerted on the fence design.

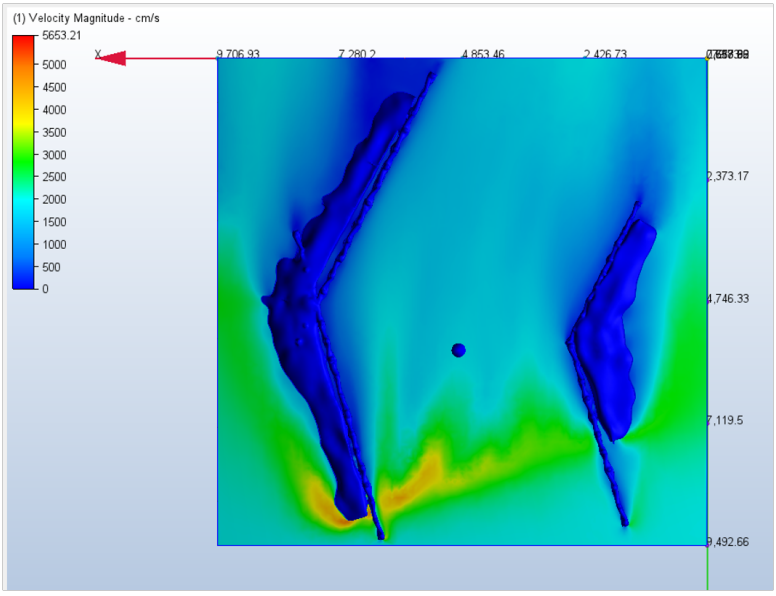


Figure 17: *Steady-state CFD simulation result showing velocity magnitude of water current in cm/s, showing areas were force exerted heavily (red) and weakly(blue)*

By analysis of the simulation results, the simulation shows that the front fence experienced significant force, indicating a potential need for reinforcement, labelled in *(Figure. 16)*as yellow to red. This

valuable information allows for informed design modifications to enhance the robustness of the fence and ensure its effectiveness in withstanding the force of the water flow in areas where the riverbanks experience higher water flow velocity.

The CFD modelling allowed for a virtual representation of an average river environment in blocking water and reducing erosion risks could be concluded. However, the steady-state modelling lacked further analysis of fluid dynamics by simulating different flow scenarios to analyse the effectiveness of the fences with varying parameters such as water velocity, pressure, and turbulence which were not taken into account under varying flow conditions in a transient-state simulation. This will further allow for a comprehensive assessment to gauge the performance of the prototype fence design. Furthermore, introducing another testing parameter, which is measuring the displacement of soil on the banks after the installation of the fences would solidify the accuracy of the testing results.

Through the CFD modelling, the simulation provided data on potential areas of erosion, and the overall performance of the fence design. These findings served as crucial input for further refinements and optimization of the prototype before its physical implementation in the river setting.

Social Impacts

This project will have impacts on the Aboriginal community of Barron River in various ways. Firstly, it will contribute to cultural preservation and empowerment by acknowledging and respecting their cultural heritage and traditions. Through engagement with community members and stakeholders, the project will incorporate elements that promote cultural preservation, fostering the community's identity, pride, and sense of belonging.

Moreover, the project will prioritise community engagement and participation, ensuring that the voices of the Djabuganjdji community are heard and their perspectives are taken into account. Through regular consultations, workshops, and collaborative partnerships, community members will actively participate in the decision-making processes. This inclusive approach will result in more culturally relevant designs and instil a sense of ownership among the community members.

Also, the project will create employment and economic opportunities for the Djabuganjdji community. By involving community members in the project's development, implementation, and maintenance, it will provide job training, skill development, and sustainable employment. This will not only enhance the socio-economic well-being of the community but also empower them to take charge of their own future.

Lastly, the project will support education and capacity building within the Aboriginal community. It will offer educational resources, training programs, and mentorship opportunities, empowering community members with knowledge and skills. This will contribute to personal growth, professional development, and overall socio-economic advancement within the community.

Economic Impacts

The economic impacts and cost estimation of the water wattle fences project would vary depending on various factors including labour costs, cost of materials, and maintenance costs.

1. Material Costs:

   Native Trees: The main materials for constructing the wattle fences are native trees such as Blackbutt and Silky Oaks, or Hazel. The cost would depend on the availability and the number of trees needed. If these trees are already available locally and can be used without causing environmental harm, the cost could be minimal.

   Rope, Twine, and Wire: These can be purchased or made from recycled materials. The cost of new rope, twine, and wire can vary, but let's estimate $200 for this expense.

Tools: Hammers, saws, nails and other required tools would also need to be accounted for. Let's budget $300 for these.

2. Labour Costs:

   Construction Labor: Depending on the size of the project, you may need a team of workers to construct and install the fences. If local volunteers are used, labour costs can be minimised. However, if workers need to be paid, labour costs could significantly increase. For example, at a rate of $20/hour for each worker, 10 workers working for 8 hours a day for a week would cost $8,000.

   Maintenance Labor: Regular maintenance would be required. Assuming that community volunteers can take on some of this responsibility, we might also consider having a paid maintenance worker. If one part-time worker is hired at $20/hour for 20 hours a week, the annual cost would be $20,800.

3. Training Costs:

   Organising training sessions for the community members is crucial. Costs associated with this might include hiring a trainer, securing a training venue, and providing necessary materials. If a local community centre can be used for training and if community members can serve as trainers, costs can be minimised. Let's estimate $500 for this.

4. Miscellaneous Costs:

   Contingency fund: It's always wise to have a contingency fund to handle any unexpected costs. This could be 10% of the total estimated costs. If we consider only the costs estimated so far, this would be around $1,080. Summing up the estimated costs:

   Materials: $500 Labour: $28,800 (construction and maintenance) Training: $500 Contingency: $1,080

   Total Estimated Cost: $30,880

Regional Economic Impacts:

- Job Creation: Implementing this project could potentially create jobs for the local community, thus improving their economic status.

- Tourism Boost: A successful project could potentially boost local tourism, thereby increasing revenue for local businesses.

- Prevention of Land Loss and Degradation: By preventing riverbank erosion, this project could save significant costs associated with the loss of land, infrastructure, agricultural productivity, and overall degradation of surrounding lands.

  Increase in Property Values: Successful erosion control could lead to an increase in property values along the Barron River, thus benefiting the local community economically.

It is highly critical to note that these are rough estimates and actual costs may vary significantly based on local conditions and factors and the availability of sourced materials. A detailed local market study and evaluation of natural resources before initiating the project is necessary.

Environmental and cultural impacts

When considering the environmental impacts of a project in the context of the Djabuganjdji community of Barron River, it is crucial to recognise the significance of the local ecosystem and the cultural connection that the community has with the land. Australian Indigenous communities often have deep-rooted relationships with their environment, which include spiritual, cultural, and sustenance aspects. Therefore, any project should take into account the potential environmental impacts and consider the unique perspectives and concerns of the Djabuganjdji community.

Biodiversity and Habitat: The Barron River area is likely to be home to a diverse range of plant and animal species. Assessing the environmental impacts involves understanding the potential disruptions to local ecosystems, such as habitat destruction, fragmentation, or alteration. These impacts can affect the flora and fauna that the Djabuganjdji community relies upon for food, medicinal purposes, and cultural and religious practices. It is essential to minimise disturbances to ensure the preservation of biodiversity and the sustainability of traditional practices.

Water Resources: The Barron River is a vital water source for both the environment and the community. Any project in the area should consider the potential effects on water quality, quantity, and flow patterns. Disruptions to water resources can impact aquatic ecosystems, affecting fish populations, water-dependent plants, and other species. Additionally, changes in water availability and quality can have direct consequences for the Djabuganjdji community's access to clean water for drinking, cultural practices, and maintaining their traditional livelihoods.

Land Use and Traditional Knowledge: Australian Indigenous communities often possess extensive traditional knowledge about the land as the Tradtional Owners, including sustainable land use practices and the protection of sacred sites. Any consequencing environmental impacts by erosion or land misuse would pose potential conflicts with traditional land uses, the design implementation approach aims to preserve and respect Djabuganjdji cultural heritage. Engaging with the community throughout the project's design and development process is a fundamental part of incorporating their perspectives and ensuring the protection of significant cultural and historical sites and the appropriateness of the project design.

Environmental Stewardship: Recognising the environmental impacts also presents an opportunity for promoting environmental stewardship and sustainability within the Djabuganjdji community. Engaging the community in conservation efforts, providing training and employment opportunities related to environmental management, and incorporating traditional ecological knowledge into decision-making processes can help foster a sense of ownership and stewardship towards the land and its resources.

In short, considering the environmental impacts of a project in the Djabuganjdji community of Barron River entails a holistic approach that incorporates biodiversity conservation, water resource management, preservation of traditional land uses, and respect for cultural heritage.

Design Implementation

The implementation plan for the water wattle fences as a sustainable solution for erosion control along the Barron River, incorporating the additional information, can be outlined as follows:

1. **Community Consultation:** Engage in a comprehensive consultation with the Djabugandji community to introduce the project, its objectives, and the community's role in its implementation and maintenance. This consultation fosters community ownership and responsibility.

2. **Material Sourcing and Preparation:** Source and prepare materials for construction, including native trees such as Blackbutt and Silky Oak, Hazel, along with locally sourced or recycled materials like rope, twine, and wire.

3. **Comprehensive Framework for Governance, Organisation, Maintenance, Training, and Funding:** This is a pivotal implementation, aiming at the sustenance and continuity of the project. A comprehensive framework for governance, organisation, maintenance, training, and obtaining funding is most essential for the successful implementation and sustainability of the project. This framework involves establishing a governing body or committee from the local Djabugandji community with defined roles and responsibilities, creating an organisational structure with clear communication channels, developing maintenance guidelines and schedules, conducting training sessions in collaboration with the community experts, and implementing a funding strategy that combines various sources of financial support. By incorporating these elements, the project can ensure effective management, community involvement, infrastructure maintenance, skill development, and sustainable funding, leading to long-term success in controlling erosion along the Barron River.

4. **Pilot Installation and Progressive Implementation:** The project will be initially implemented as a pilot installation in a high erosion area to study the design and assess any needed modifications. Based on the success of the pilot installation and the availability of funding, the installation will then be carried out in progressive stages, involving community members, project teams, and volunteers in the construction and installation of the water wattle fences along the riverbanks, prioritising areas of high erosion. This fosters community involvement and a sense of ownership. This phased approach allows for continuous evaluation and adaptation of the project while ensuring its scalability and sustainability.

5. **Training and Education:** Conduct training sessions in both English and the Djabugay language to educate community members on fence maintenance and potential construction. Involve Indigenous community members in organising and delivering these sessions.

6. **Ongoing Maintenance:** Establish a system for regular maintenance checks by designated community members or volunteers to ensure the longevity and effectiveness of the fences. Inspect for wear, damage, and potential hazards caused by natural factors.

7. **Funding:** Secure funding through a combination of petitions for governmental support, with the endorsement of local Djabugandji elders and local Queens-land council members' governmental support, community donations, and non-profit organisations. Incorporation of social media to promote the project to garner awareness and obtain donations with the aid of crowdfunding platforms such as GoFundMe can be vital monetary resources too to ensure long-term sustainability.

8. **Timeline:** The implementation timeline will be determined based on resource availability, successive stages of implementation and design refinement after pilot implementation, available funding, community participation, and environmental factors. Flexibility may be required to accommodate these variables, and the execution will be done based on the success re-assessment of the implementation of the fences and the erosion system sustenance as a whole.

By following this implementation plan, the project can effectively control erosion while actively involving the Djabugandji community. This approach ensures the project's sustainability and longevity holistically, as well as the empowerment of the community in addressing environmental challenges along the Barron River.

Conclusion

This report presents a sustainable solution for erosion control along the Barron River near Dungarra, Australia, focusing on the use of water wattle fences. The chosen design effectively protects mangroves, is easy to maintain by the community, and aligns with the values of the Djabugandji community.

The implementation plan includes a comprehensive governance and organisational framework, ensuring the long-term sustainability of the project. This framework outlines guidelines for governance, organisation, maintenance, training, and obtaining funding. By establishing clear protocols and responsibilities, the project can continue to thrive and benefit the community.

While the proposed design was tested using a 3D-printed prototype and CFD simulation has confirmed the final design's efficacy to a preliminary degree to conduct a pilot implementation, however, to improve the accuracy of the testing results it is advisable to obtain additional information that is needed about the real-life geographical data of the river including river morphology and terrain, and more accurate data of water flow velocity conditions in various climatic conditions and locations, further simulation should be conducted incorporating these variables in a transient-state simulation rather than a steady-state static simulation which the study would significantly improve the design robustness. Alternatively, testing in real-life conditions would provide also a more realistic approach assessment of the design's performance, which will be part of the pilot implementation. The limitation of available data about the river conditions and morphology hinders a complete understanding of the project's impact. Therefore, gathering more data and conducting thorough assessments of the river's characteristics would strengthen future implementations.

In conclusion, the implementation of water wattle fences and the implantation of mangrove seedlings, supported by a comprehensive governance and organisational framework, offers economic, social, and environmental benefits. The solution aims to ensure alignment with the values of the Djabugandji community, acceptance and stewardship by the community, and protection of the river's ecosystem and ensuring its long-term sustainability. To improve the project, further testing in real-life scenarios and acquiring more data about the river conditions and morphology are recommended.

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